Question		Answer	Marks	AO	Guidance
17		**DR**			
		$[\bar{x} =]347.34$ seen	**B1**	**1.1**	**BC** allow 347.3 or 347; allow eg mean or $\bar{X}$, but no other symbol if seen
		$\left(s^2 = \frac{6036597-50\times347.34^2}{49}\right) = 88.637$ or $(s =)9.4147$ seen **BC**	**B1**	**1.1**	**NB** 88.6371428571; may see 9.4147(3010007) ; accept awrt 88.6 or awrt 9.41; do **not** allow 86.86 or 9.320; allow eg σ^2 or VAR[X]
		$H_0: \mu = 350$ $H_1: \mu < 350$	**B1**	**1.1**	allow use of other parameters (but not $\bar{x}$ or $\bar{X}$) if defined as [population] mean mass; allow if hypotheses stated in words: may earn **B1B0** if "population" omitted
		μ is the **population mean mass** of honey in a jar	**B1**	**2.5**	condone weight instead of mass
		$N\left(350, \frac{\text{awrt } 88.6}{50}\right)$ **oe seen**	**B1**	**3.3**	**FT** their s^2 correct to 3 sf or better; may see N(350, 1.77(274285741) or $N\left(350, \frac{9.415^2}{50}\right)$ or $N(350, 1.331^2)$
		$(P(\bar{X} < 347.34) =)$ awrt 0.023	**M1***	**3.4**	allow **M1** for correct probability found from $N\left(350, \frac{\text{their awrt } 88.6}{50}\right)$; must see their distribution if value incorrect; condone slip in notation eg use of $P(X < 347.34)$; allow for use of 347 or 347.3 instead of 347.34
		their 0.02287 correctly compared with 0.01	**M1dep***	**3.4**	allow eg 2.28% > 1%
		do not reject H_0 **or** accept H_0 **or** reject H_1 or not significant	**A1FT**	**1.1**	**A0** if their 0.0229 < 0.01
		there is **insufficient evidence** at the 1% level to **suggest** that the **mean mass** of honey in a jar is **less than** 350 grams	**A1**	**2.2b**	do not allow eg conclude / prove / indicate or other assertive statement instead of suggest; this mark is dependent on the award of all preceding marks apart from the fourth and fifth **B** marks ; condone weight instead of mass; must come from correct probability: $(P(\bar{X} < 347.34) =)$ awrt 0.023

Question		Answer	Marks	AO	Guidance
		Alternatively, for the last 4 marks critical region is $\bar{X} < 346.9(0)$ or critical value is $\bar{X} = 346.9(0)$ to 3 or more sf	**M1***		allow **M1** for critical region found from $\text{N}\left(350, \frac{\text{their awrt } 88.6}{50}\right)$; must see their distribution if value incorrect; condone slip in notation eg use of X
		347.34 correctly compared with their 346.90	**M1dep***		allow 347 or 347.3 compared with 346.9(0)
		do not reject H_0 **or** accept H_0 **or** reject H_1 or not significant	**A1FT**		**A0** if their 347.34 < their 346.9(0)
		there is **insufficient evidence** at the 1% level to **suggest** that the **mean mass** of honey in a jar is less than 350 grams	**A1**		do not allow eg conclude / prove / indicate or other assertive statement instead of suggest; this mark is dependent on the award of all preceding marks apart from the fourth and fifth **B** marks; must have correct critical region
		Alternatively, for the last 4 marks $z\left[=\frac{347.34-350}{\frac{9.414}{\sqrt{50}}}\right] = \text{awrt} -1.998$	**M1***		allow **M1** for z found from $\text{N}\left(350, \frac{\text{their awrt } 88.6}{50}\right)$; must see their distribution if value incorrect; allow for use of 347 or 347.3 instead of 347.34 NB may see $\text{awrt} -2.03$ or $\text{awrt} -2.25$ from 347.3 or 347
		their $-1.998 > -2.33$	**M1dep***		their z correctly compared with -2.326347877 ...to 2 or more dp
		do not reject H_0 **or** accept H_0 **or** reject H_1 or not significant	**A1FT**		**A0** if their $-1.998 < -2.33$
		there is **insufficient evidence** at the 1% level to **suggest** that the **mean mass** of honey in a jar is less than 350 grams	**A1**		do not allow eg conclude / prove / indicate or other assertive statement instead of suggest; this mark is dependent on the award of all preceding marks apart from the fourth and fifth **B** marks; z must be correct
			[9]		